

Football Analytics Platform

UBC MDS Capstone 2026 — Trilemma Foundation

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Problem & Roadmap

The gap: StatsBomb event data exists, but analysts need complex SQL to use it.

RQ	Question	Who
RQ1	What tactical archetypes exist across competitions?	Li · clustering + dashboard
RQ2	Do players perform consistently club vs nationally?	Rabin · Consistency Explorer
RQ3	Can analysts query football data in plain English?	Teem · chatbot
—	How does data get there?	Quan · dbt + Dagster

Data Pipeline — dbt

StatsBomb API → GCS → BigQuery → **dbt** (3 layers) → chatbot + dashboard

- **Staging** — type casts only, 1:1 mirror
- **Intermediate** — 270-min filter, per-90 metrics, men's only
- **Marts** — ML outputs joined in

dbt = **Makefile + git for SQL** — version-controlled `SELECT` s with tests.

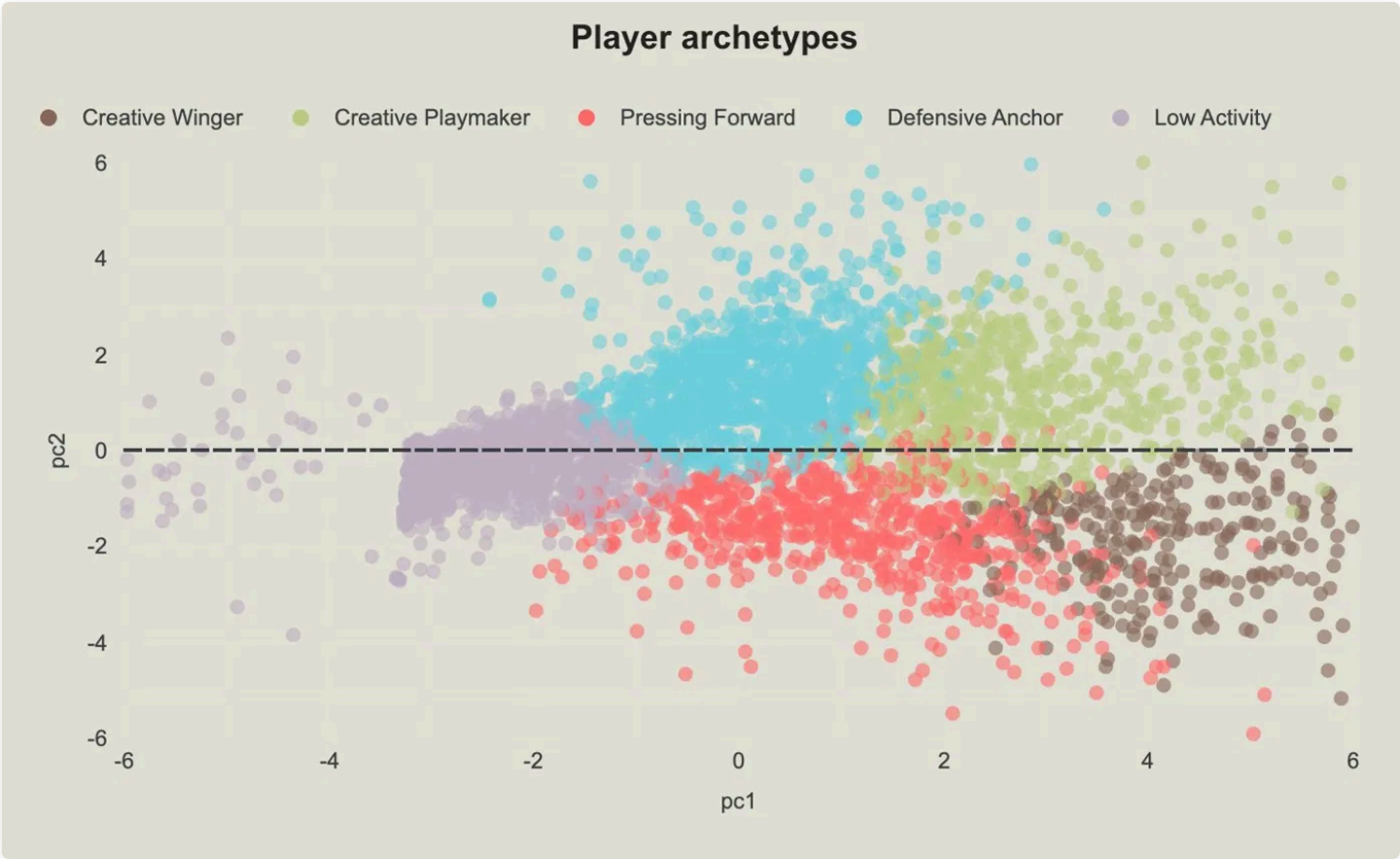
*Orchestrated by **Dagster** — next slide.*

Pipeline Orchestration — Dagster

Dagster = cron + Makefile dependency graph (like GitHub Actions job order)

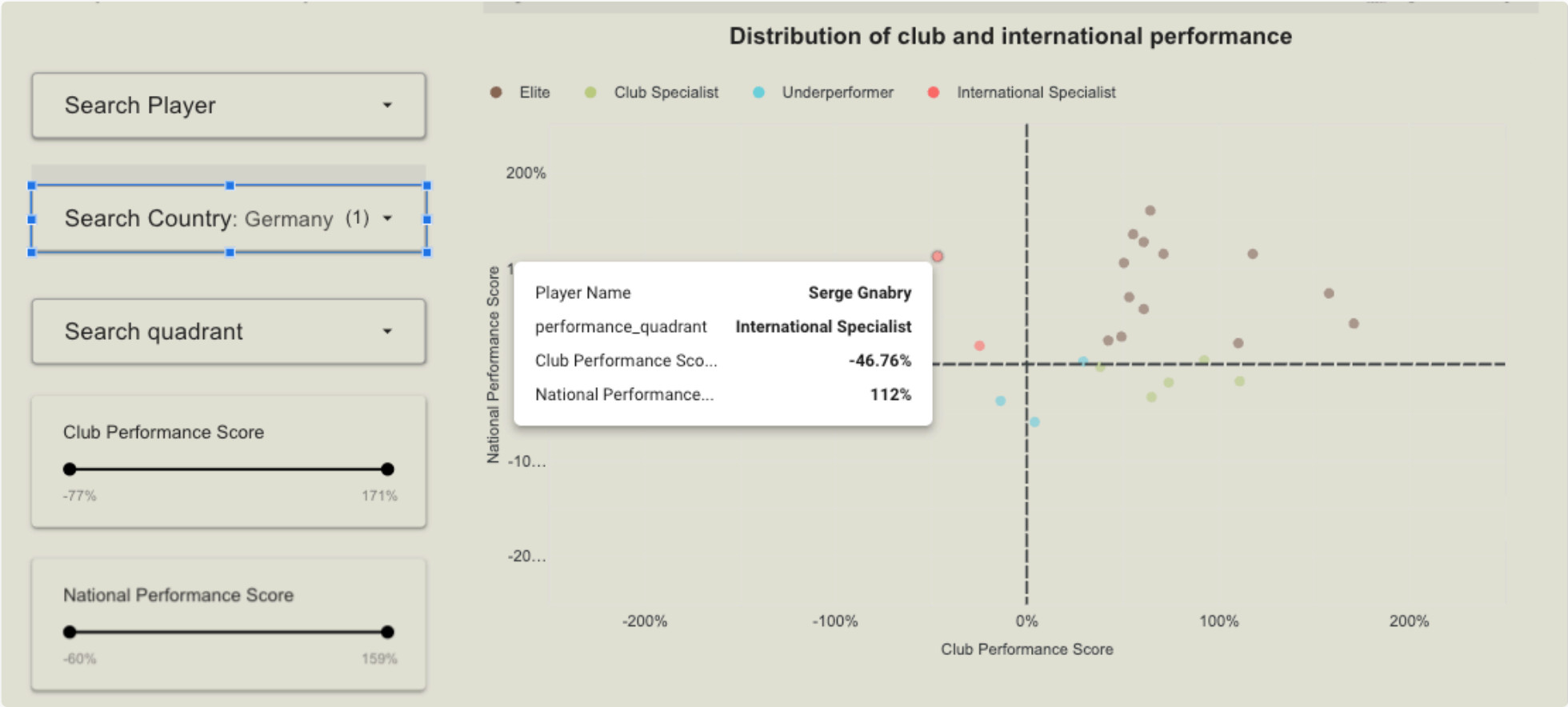
- **Weekly** — full pipeline, Monday 6:00 AM UTC
- **Hourly sensor** — new GCS file → dbt + ML + marts only
- Hands-off for Trilemma — no manual runs

RQ1: 5 Tactical Archetypes Found Across 15 Competitions

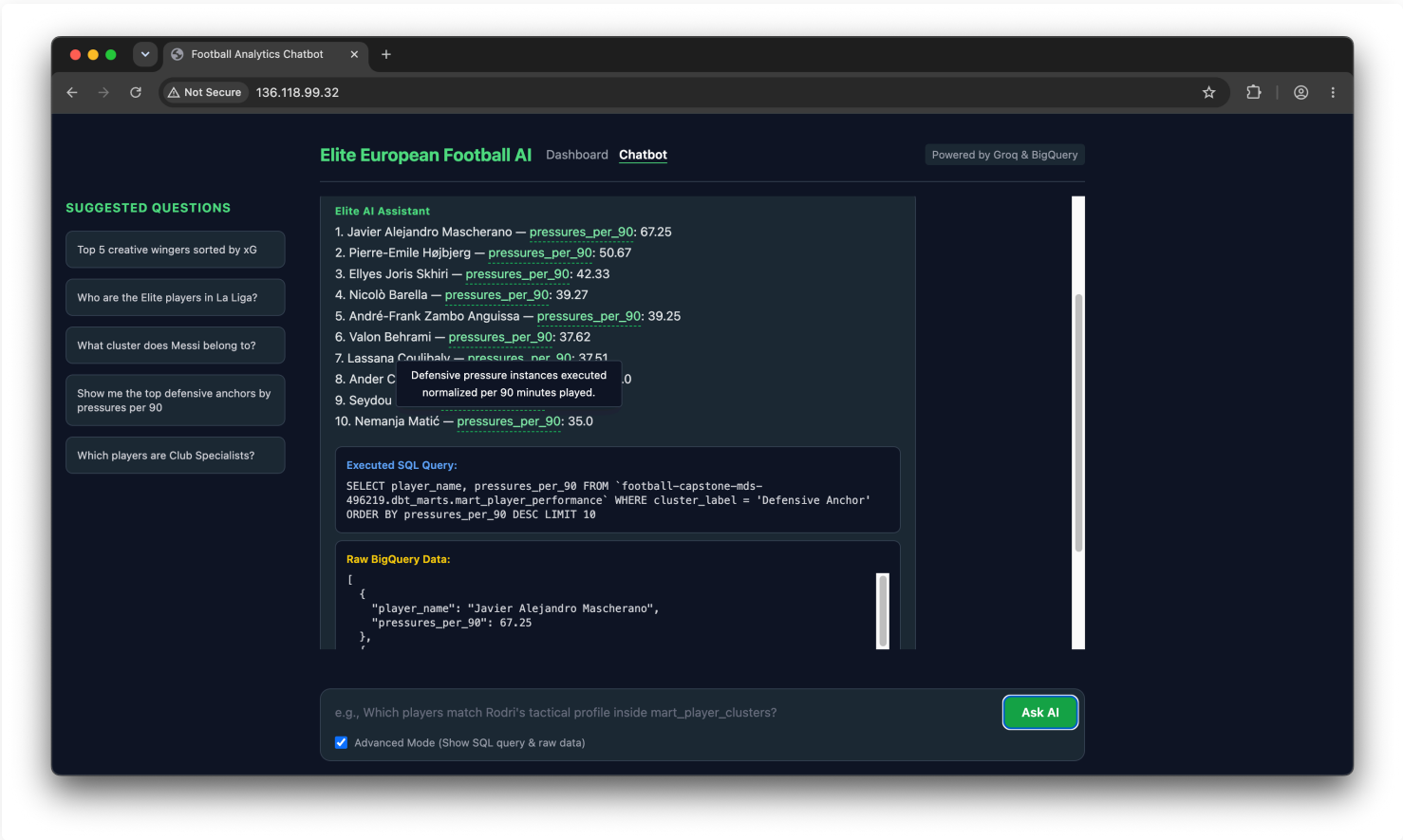


5 archetypes · 5,800+ player-seasons · 15 competitions

RQ2: Most Players Perform Differently Club vs Nationally



RQ3: Any Football Question, Answered in Plain English



Plain English in · answers in < 10 sec · no SQL

Clustering Pipeline

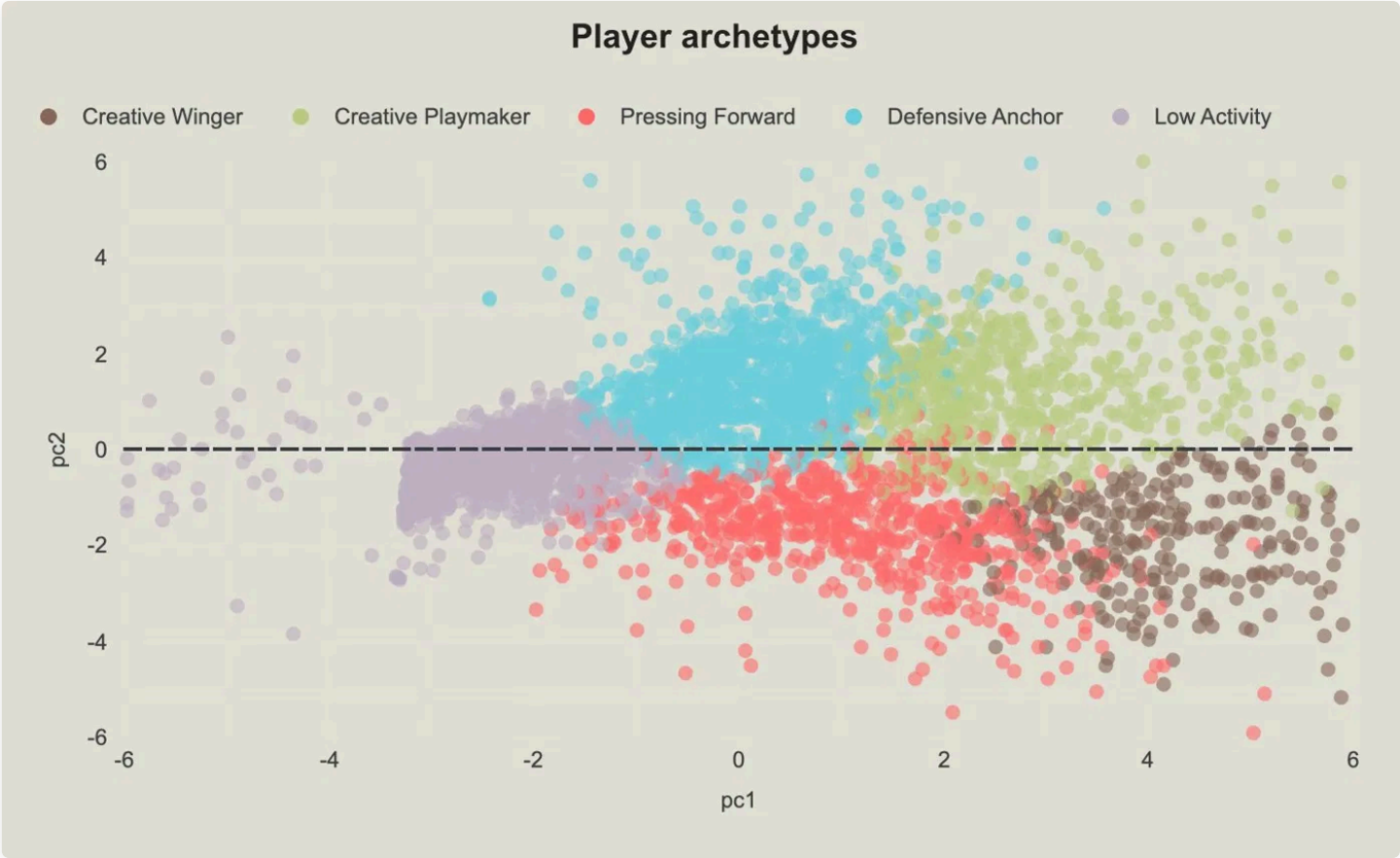
RQ1: What player archetypes exist across European football?

Step	Detail
Input	<code>int_player_season_stats</code> from dbt
Filter	≥ 270 min · goalkeepers excluded
Features	13 per-90 metrics (xG, passes, pressures, ...)
Model	Median impute · clip outliers · Scaler → PCA → KMeans k=5
Output	5 archetypes → <code>cluster_assignments</code> → Looker + chatbot

Cluster Results: 5 Archetypes

Archetype	Style
Creative Winger	xG, dribbles, att. third passes
Creative Playmaker	Passes, carries, chance creation
Pressing Forward	Pressures, shots, defensive actions
Defensive Anchor	Interceptions, clearances, blocks
Low Activity	Below-cohort activity

Cluster Scatter — Looker Studio



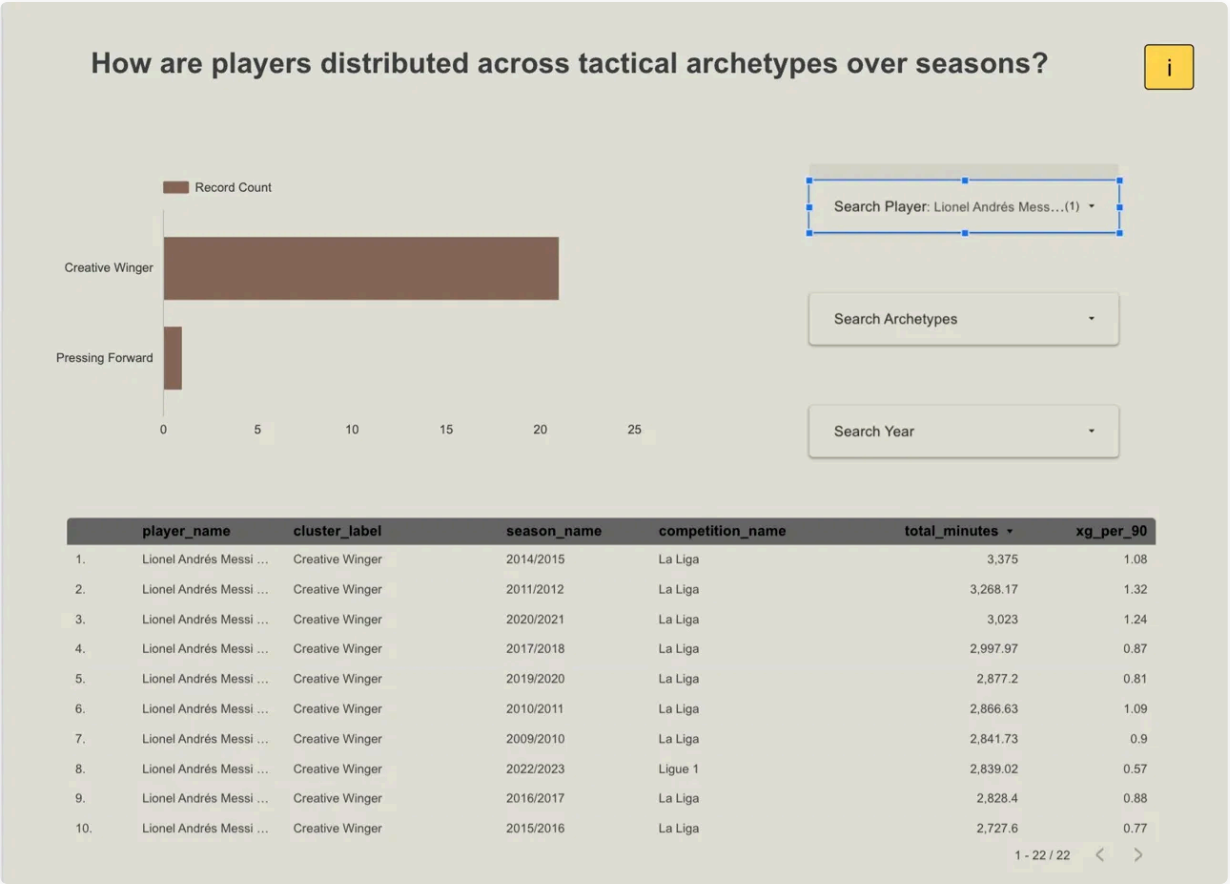
PCA scatter · 5,759 player-seasons · 5 archetypes · PC1 = activity level · PC2 = attacking vs defensive profile

Player Archetypes Dashboard — Overview



5,759 player-seasons · 15 competitions · Low Activity and Defensive Anchor are the largest groups · filter by archetype, year, or competition

Player Archetypes Dashboard — Explorer



Messi classified as Creative Winger across 22 La Liga seasons · search any player to see their archetype history

Consistency: The Question & Pipeline

RQ2: Do players perform differently for club vs national team?

```
int_player_club_vs_national → consistency.py → consistency_scores → Looker Explorer
```

- **Eligibility:** ≥ 270 minutes in **both** contexts (~3 full matches)
- **Cohort:** 493 dual-context players across **58 countries**
- Z-scores computed **within each context** (club peers vs club, national vs national)

Consistency Formulas & Results

Formulas

1. $z = (x - \mu_{\text{context}}) / \sigma_{\text{context}}$ within each context
2. **Performance score** = $\sum (z \times \text{PCA weight})$ — club & national axes on scatter plot
3. **Consistency score** = $1 - \text{mean}(|z_{\text{club}} - z_{\text{nat}}|)$ — profile similarity, not quality

Results

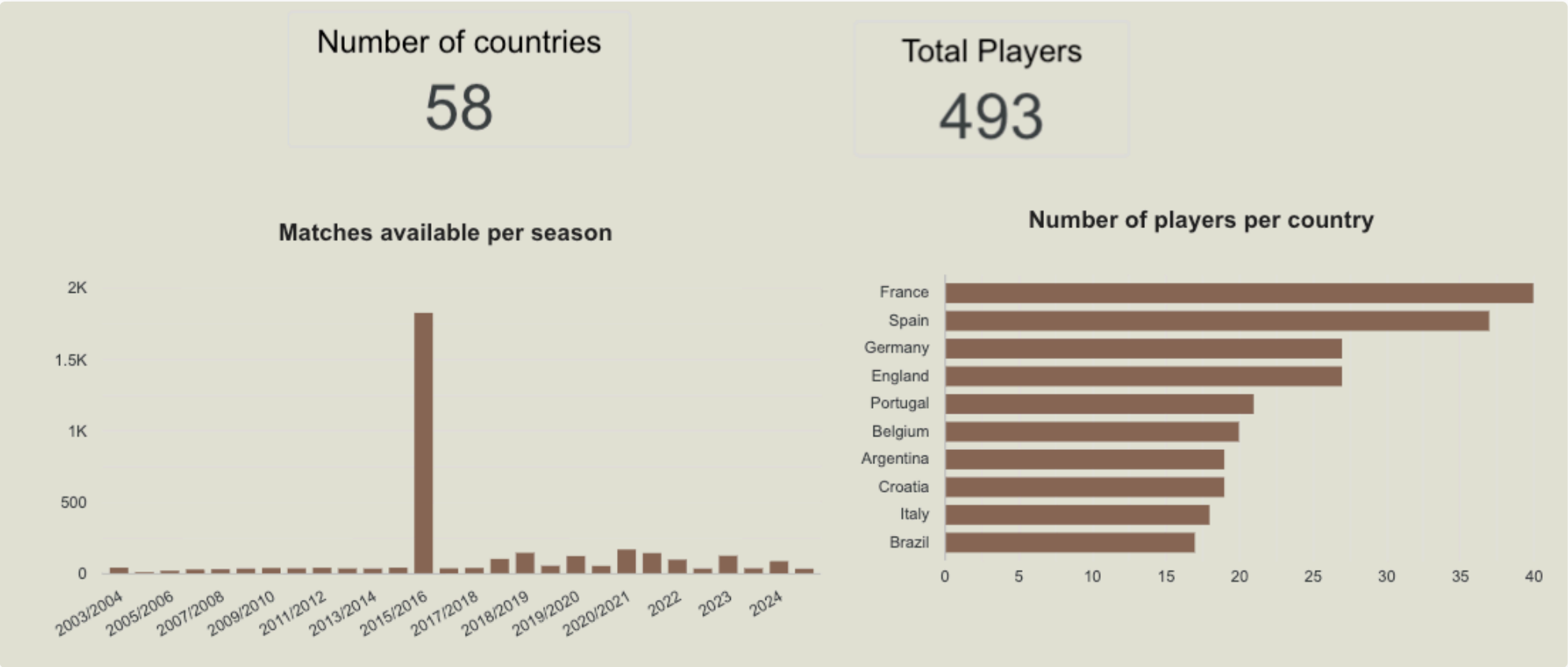
- **493 players** · **58 countries** · ~25% per quadrant
- **Most consistent** → profile barely changes between contexts
- **Germany** leads national score · **Sweden** lowest in cohort

Quadrant Definitions

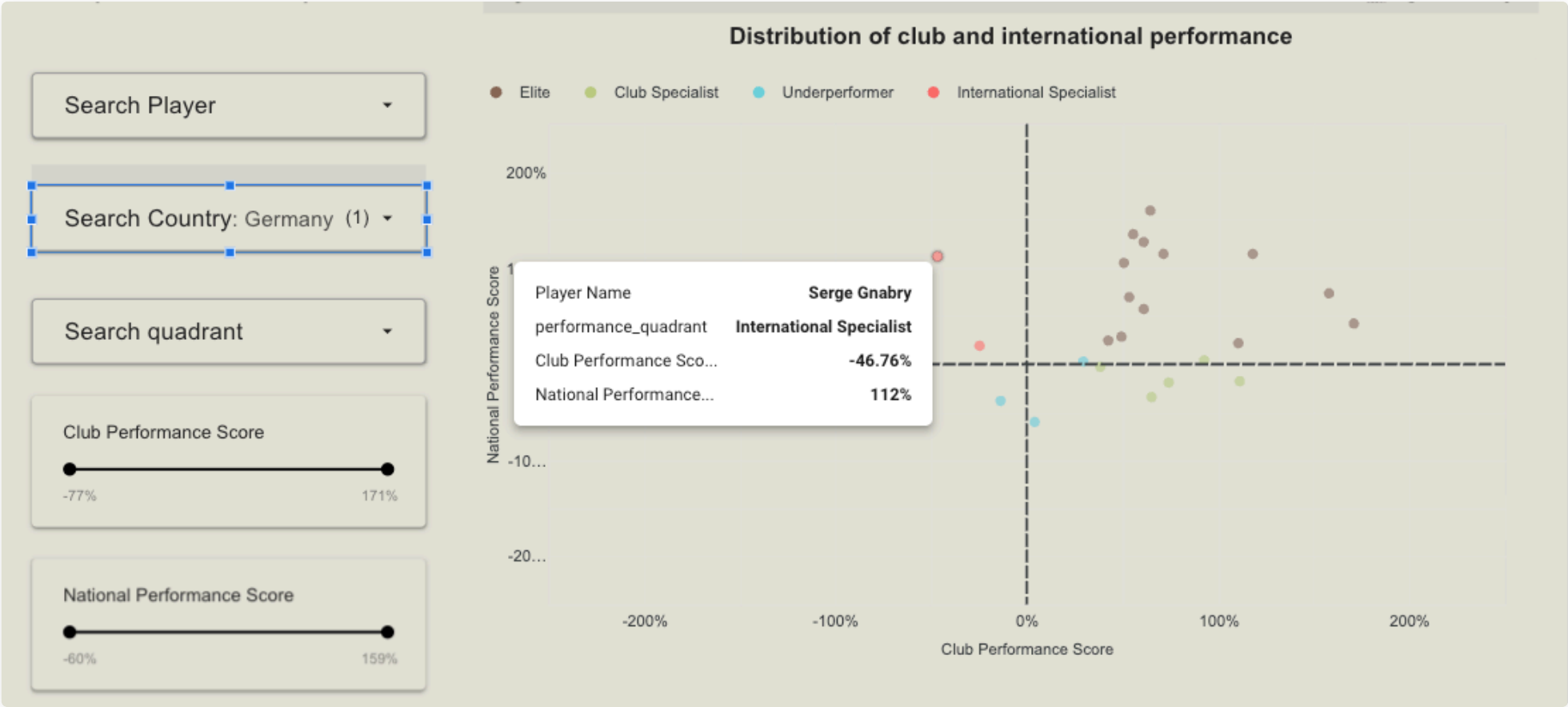
Split cohort at **medians** of club & national performance (~25% each)

Quadrant	Rule
Elite	$\geq$ median in both contexts
Club Specialist	Club $\geq$ median, national $<$ median
International Specialist	Club $<$ median, national $\geq$ median
Underperformer	Below median in both

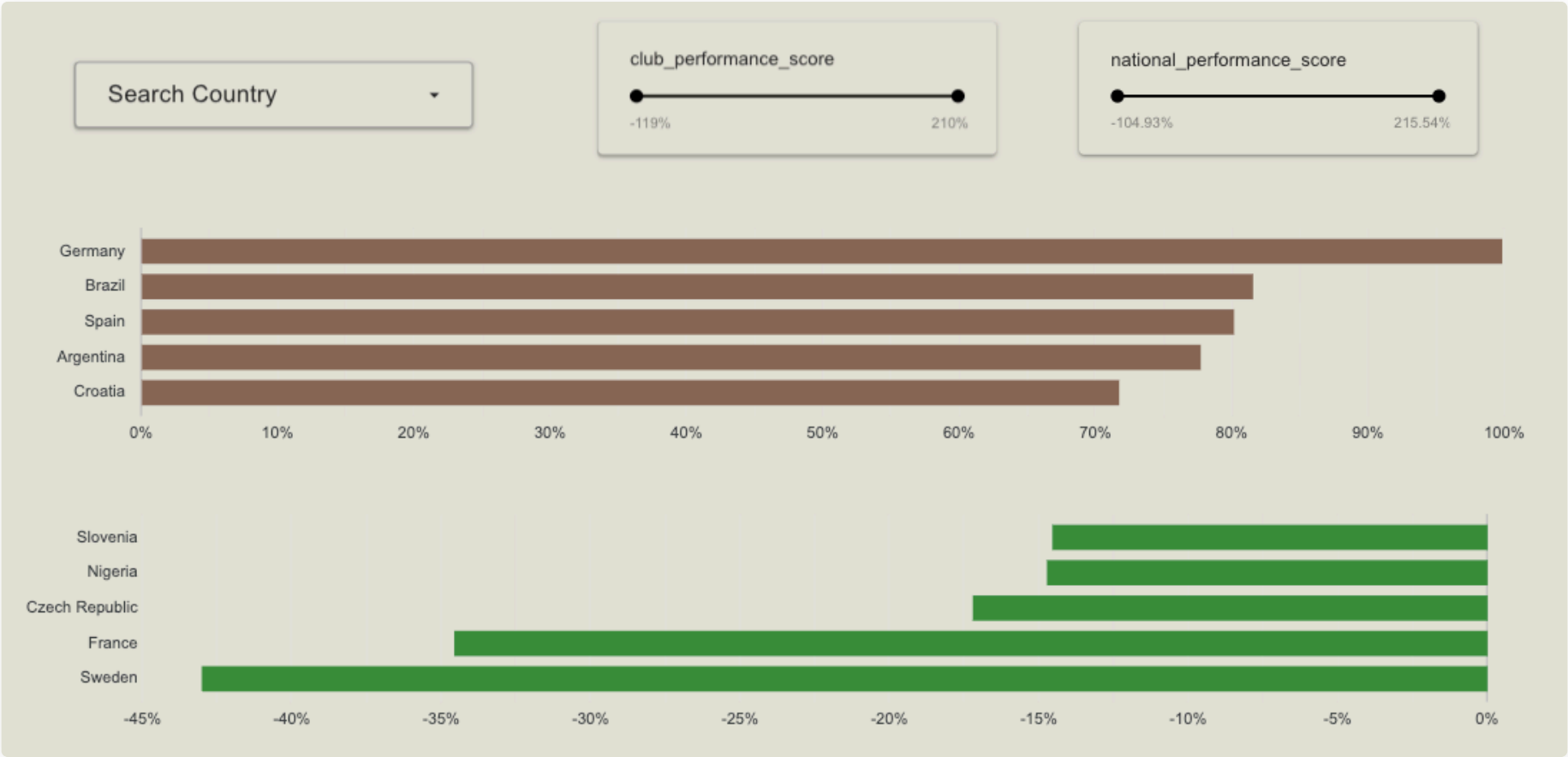
Consistency Explorer — Overview



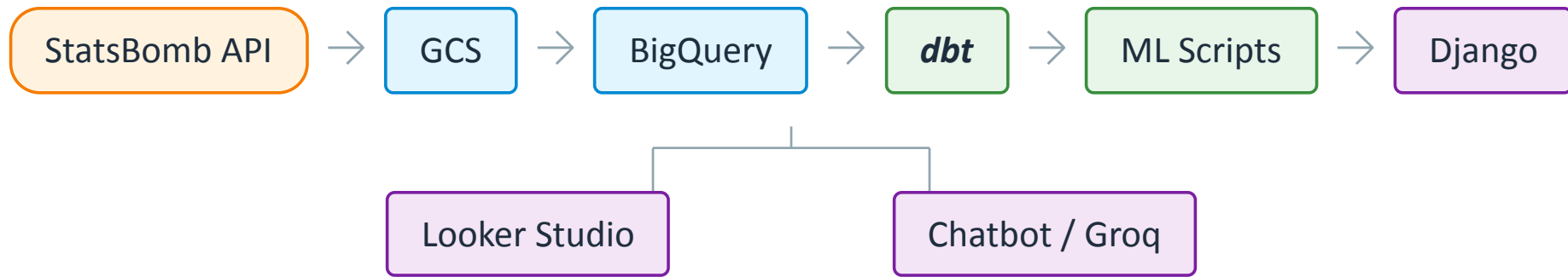
Consistency Explorer — Quadrant View



Consistency Explorer — Country Rankings



System Architecture



End-to-end pipeline: data → serving

Architecture — Components

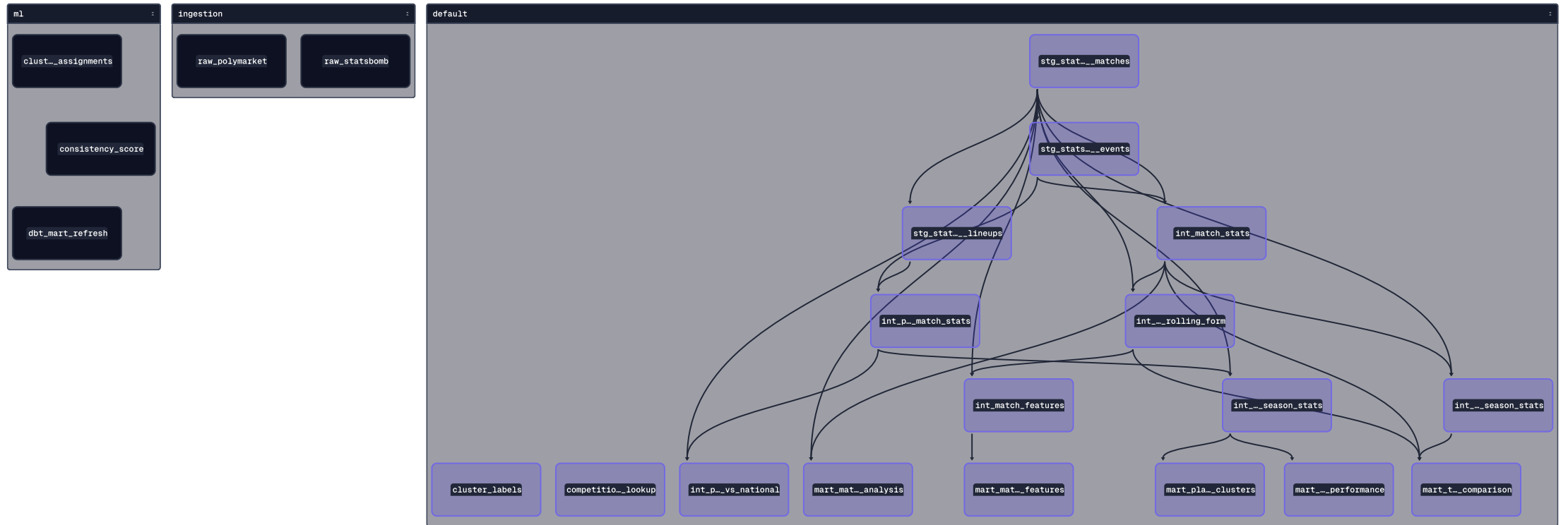
- **Ingestion:** StatsBomb API → Google Cloud Storage (GCS)
- **Warehouse:** Centralized in Google BigQuery
- **Transformation:** 3-layer dbt architecture
- **Machine Learning:** Python scripts for PCA/KMeans and Consistency Scoring
- **Serving:** Django web app · Looker Studio dashboard · LLM chatbot

dbt Layers (Detail)



- **Staging** — 1:1 mirror, type casts only
- **Intermediate** — men's filter, 270-min threshold, per-90 metrics
- **Marts** — ML outputs joined in → chatbot + Looker read here

Dagster Asset Graph

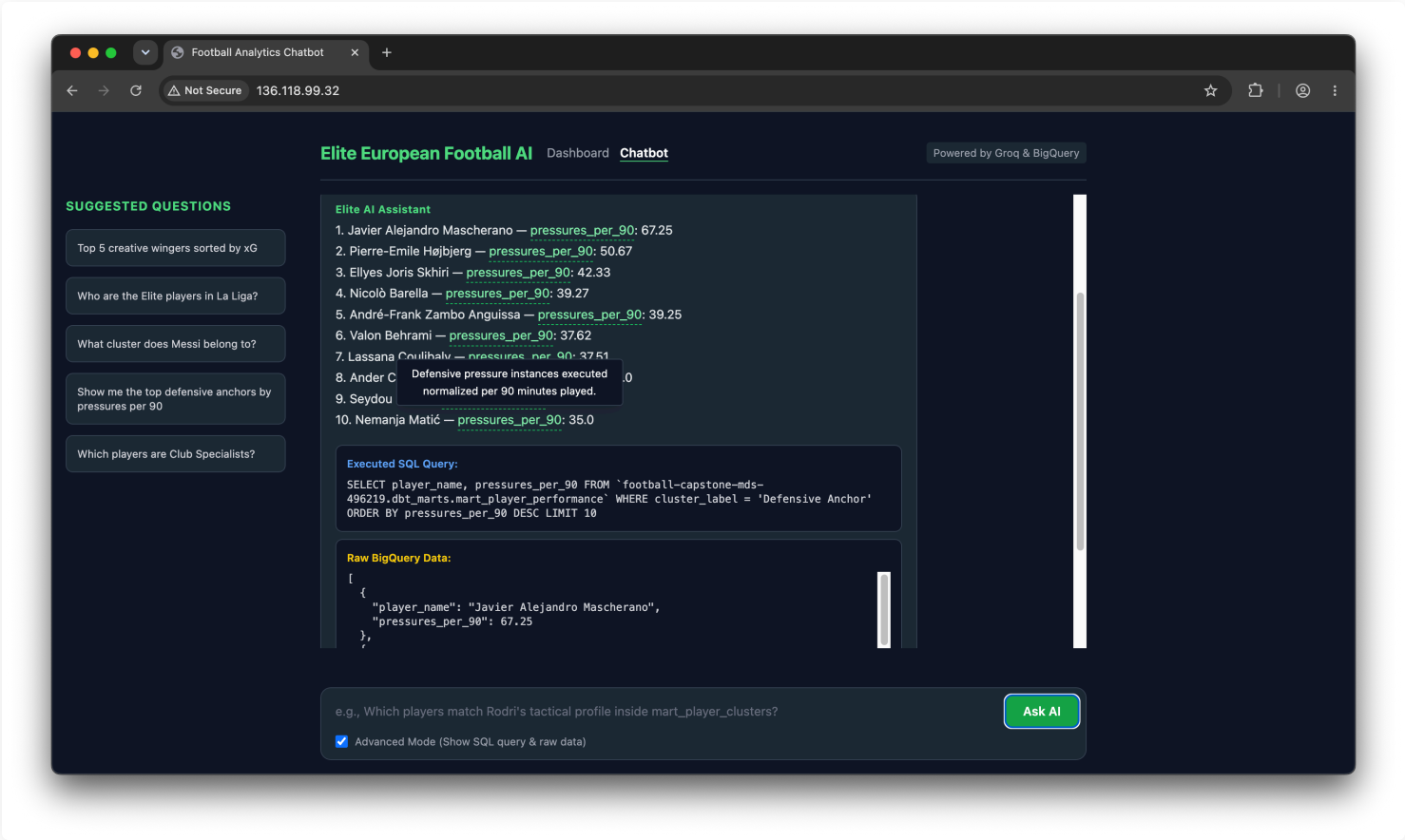


Asset graph · ingestion → dbt → ML → marts

Dagster — How It Runs

- **Asset graph** — run in order: Ingestion → dbt → ML → mart refresh
- `full_pipeline_job` — every Monday 6:00 AM UTC
- **Hourly GCS sensor** — new match file → `post_ingestion_job` only (no re-ingestion loop)

Chatbot Demo



Plain-English question → BigQuery answer

Chatbot — How It Works

- **Engine:** Groq `llama-3.3-70b-versatile`
- **Tool-calling:** English → BigQuery SQL → formatted answer
- **Security:** `is_safe_sql` blocks non-SELECT statements
- **Advanced mode:** shows generated SQL for debugging

Limitations & Future Scope

- **Goalkeepers** — excluded from PCA clustering (`cluster = -1` , archetype Goalkeeper)
- **Polymarket odds** — manual GCS upload only; no automated ingestion yet
- **Women's football** — men's filter at intermediate layer; needs parallel dbt models

Partner Handover — What Trilemma Gets

- Fully documented **GitHub repository**
- **Quarto** final report
- `HANDOVER.md` — component runbooks
- `docker compose up` → Django app + Dagster UI

Partner Handover — Repository

Partner Handover Guide

UBC MDS Soccer Capstone 2026 — Trilemma Foundation Football Analytics

Last updated: June 2026 **Outgoing team:** Quan Hoang, Li Pu, Teem Kwong, Rabin Duran **For questions contact:** [quandothoang \(GitHub\)](#)

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Thank You

Questions?

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